LEARNER GUIDE

Sine and cosine

Introduction to Trigonometry. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

In Lesson 3 you met the first named ratio, the tangent. This lesson completes the set. Alongside the tangent there are two more ratios — the sine and the cosine — and this time they bring in the hypotenuse. The sine of an angle is the opposite divided by the hypotenuse; the cosine is the adjacent divided by the hypotenuse. All three are captured by a single memorable map, SOH · CAH · TOA. You explore how sine and cosine change as the angle changes (as one rises, the other falls, and both stay between 0 and 1), you practise naming which ratio a pair of sides gives, and you discover a quiet piece of symmetry: the sine of one acute angle equals the cosine of the other. The question you are exploring is: how do the other two ratios complete the picture?

What you are learning to notice

- That $\sin\theta$ = opposite ÷ hypotenuse (SOH) and $\cos\theta$ = adjacent ÷ hypotenuse (CAH), while $\tan\theta$ = opposite ÷ adjacent (TOA).
- That sine and cosine both use the hypotenuse, but the tangent does not.
- That as the angle grows, $\sin\theta$ increases towards 1 while $\cos\theta$ decreases towards 0 — and both always sit between 0 and 1.
- That the sine of one acute angle equals the cosine of the other, because they add to 90°: $\sin\theta = \cos(90° - \theta)$.

Moving through it, one step at a time

In Ratio Trio, press $\sin\theta$, $\cos\theta$ or $\tan\theta$ on a 3–4–5 triangle: the two sides that ratio uses light up, and its value appears as a fraction — SOH, CAH, TOA. In Ratio Explorer, drag the angle while the hypotenuse stays a fixed length; watch the opposite grow and the adjacent shrink, and watch $\sin\theta$ rise as $\cos\theta$ falls. In Which ratio is it?, two sides are highlighted and you decide whether they give sine, cosine or tangent. In The other angle's cosine, you work out the sine of one acute angle and the cosine of the other and find they are equal. Use the Scratchpad to draw a triangle and work out all three ratios for yourself.

> **A gentle reminder**
> You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (naming the opposite, adjacent and hypotenuse relative to a marked angle (Lesson 2), the tangent ratio $\tan\theta$ = opposite ÷ adjacent (Lesson 3), and a ratio written as a division), open a short recap and return whenever you are ready.

- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Learner Guide v0.1.